MD TANVIR HASAN

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PROFESSIONAL SUMMARY

ML/AI Engineer with project-based experience developing machine learning, NLP, computer vision, and Retrieval-Augmented Generation (RAG) systems. Skilled in Python, PyTorch, Scikit-learn, LangChain, Hugging Face, FastAPI, vector databases, Docker, Kafka, PostgreSQL, and cloud technologies. Experienced in building end-to-end ML pipelines, optimizing model inference, developing APIs, and deploying AI applications.

TECHNICAL SKILLS

Programming: Python, SQL **Machine Learning:** Scikit-learn, PyTorch, Deep Learning, NLP, Computer Vision
AI/LLM: LangChain, Llama 3, Hugging Face, FinBERT, RAG, Embeddings, Vector Search
Data & Infrastructure: Qdrant, PostgreSQL, Kafka, Docker, FastAPI, GCP
Computer Vision: OpenCV, YOLOv8, Object Detection, ONNX Runtime, FP16 Quantization
Tools: Git, GitHub, Jupyter, Streamlit, MS Office

EXPERIENCE

Machine Learning Engineer (Project-Based / Independent) **Jan 2025 – Present**

- Developed ML and LLM pipelines using Python, PyTorch, and cloud infrastructure for machine learning applications.
- Optimized deep learning models using quantization and batch inference techniques to improve inference efficiency.
- Built RESTful APIs with FastAPI to integrate machine learning models with application interfaces and dashboards.

PROJECTS

Enterprise RAG & Document Intelligence System
Technologies: Python, PyTorch, LangChain, Qdrant, FastAPI, Docker, Llama 3

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) pipeline for querying multi-page PDF documents using hybrid dense retrieval and BM25 search.
- Implemented dynamic Cross-Encoder re-ranking and containerized the application with Docker, exposing inference endpoints through FastAPI.
- Reduced document retrieval latency by 45% to below 250ms and achieved 92% accuracy on benchmark testing data.

Real-Time Financial Market Sentiment Pipeline
Technologies: Python, Hugging Face (FinBERT), Scikit-learn, Kafka, PostgreSQL, Streamlit

- Fine-tuned FinBERT for financial market sentiment classification and generated structured sentiment metrics from financial text.
- Built a Kafka-based streaming pipeline to ingest news data and store sentiment metrics alongside asset price data for quantitative analysis.
- Achieved an 89% F1-score and presented real-time sentiment metrics through an interactive Streamlit dashboard.

Computer Vision Anomaly & Defect Detection System
Technologies: OpenCV, PyTorch, YOLOv8, ONNX Runtime, GCP

- Designed and trained an object detection model to identify visual anomalies in real-time video feeds.
- Optimized model inference by exporting PyTorch weights to ONNX format and applying FP16 quantization.
- Increased inference speed by 3.2× to 60 FPS while maintaining 94.5% mAP@0.5 accuracy.

EDUCATION

Diploma in Medical Technology – Laboratory Medicine **2021–2022**
Ahsania Mission Institute of Medical Technology, Dhaka | Board: The State Medical Faculty

Secondary School Certificate (SSC) – Science **2021**
Salupahloan High School | Board: Dhaka | GPA: 4.28/5.00

CERTIFICATIONS

AWS Certified Machine Learning – Specialty Amazon Web Services
Deep Learning Specialization DeepLearning.AI